SRM TRP Engineering College
(An AUTONOMOUS Institution)
Tiruchirappalli

Higher Order Assessment Instructions with Rubrics

Department	Computer Science and Engineering										
Programme:	B.E.		**Semester:**	I		**AY:**	2026 – 2027 [ODD]				
Faculty Handling:	Mr. A. Sendhooran, AP/CSE										
Course Code & Title:	2611CSSJ102 & Web Essentials										
Assessment Task:	Mini Project (Project Based Learning)		**Assessment Type:**	Term Work / Individual			**Required Hrs:**	2 Weeks			
Assessment Question based on:	*Open Ended*	✓	*Problem Based Learning*		*Creative Thinking*	✓					
	Close Ended		*Project Based Learning*	✓	*Critical Thinking*	✓					
	Any Other Type	MCQ or any other	*Challenge Based Learning*		*Level of Thinking (RBT)*	K3/K6					
SDG to be Addressed:	SDG is Required	Yes	*SDG Number (Specify or Mention Students Choice)*	*SDG 4 / 9*	*Target Number (If Applicable)*	*4.4 / 9.c*					
SDG Integration Level: **(If Addressed)**	*Level 1: Awareness*		*Level 2: Design Alignment*	✓	*Level 3: Impact and Measurement*						
RIASEC Context	Realistic ✓	Investigative ✓	Artistic ✓	Social	Enterprising	Conventional					
Presentation Format:	Live Demonstration (localhost) / PDF Report / GitHub Repository										
Max Marks:	**40 Marks**										
Relevant Course Outcome(s):	CO1, CO2										

Submission Deadline:	**09/10/2026**
Student Name & Register No.	J. Apsaras / TRP26CS021
Allocated Website Topic	Department of Master of Computer Application (MCA) Website

1. Topic, Aim and Objectives

Course: 2611CSSJ102 – Web Essentials (HTML5 and CSS)
Topic: Department of Master of Computer Application (MCA) – informational website
Aim: To design and develop a simple, responsive multi-page website for an MCA department. The website introduces the programme, lists representative courses and software tools, showcases sample student projects, and provides a registration form and contact information.

Objectives

- Create connected HTML5 pages with clear navigation and semantic structure.
- Present MCA courses in an accessible table and describe common software tools.
- Display sample software projects and their purpose.
- Use HTML lists, hyperlinks, tables, forms and suitable input types.
- Apply basic CSS to improve readability and support mobile screens.
- Run the site locally, test links and form validation, and document the result.

2. Components Used

2.1 Hardware and Software

Component	Specification / Purpose
Computer	PC or laptop with a modern operating system
Code editor	Visual Studio Code or any text editor
Web browser	Google Chrome, Microsoft Edge or Firefox
Local web server	Python http.server or XAMPP Apache
Validation	W3C Nu HTML Checker
Version control (optional)	Git and GitHub for storing source code

2.2 HTML5 and CSS Features

Area	Elements / features	Purpose
Structure	header, nav, main, section, article, footer	Organise page content
Text and lists	h1–h3, p, ul, strong	Present information clearly
Tables	table, caption, thead, tbody, th, td	Show course details
Forms	form, label, input, select, textarea, button	Collect sample registration details
Input types	text, email, tel, date	Provide suitable data entry controls
CSS	selectors, colours, flexbox, spacing, media query	Style and adapt pages to screen size

2.3 Project Folder

```
mca-department/
  index.html
  courses.html
  software.html
  projects.html
  register.html
  contact.html
  style.css
```

All files are kept in one folder so relative links such as `courses.html` and `style.css` work when the website is opened locally.

2.4 Procedure

1. Install Visual Studio Code, a web browser (Chrome, Edge or Firefox) and Python (needed only for the local server).
2. Create a folder named `mca-department` and open it in Visual Studio Code.
3. Create `style.css` and add the basic styles: page reset, fonts and colours, header and footer, navigation bar, tables, form controls and the small-screen media query.
4. Create `index.html` with the HTML5 skeleton (`<!DOCTYPE html>`, `html`, `head`, `body`). Link the stylesheet with `<link rel="stylesheet" href="style.css">` and add the viewport meta tag for mobile screens.
5. Add the semantic structure: `header` with the department title and the `nav` links, a `main` area with sections and a list of highlights, and a `footer`.
6. Create `courses.html` and build the course table using `caption`, `thead`, `tbody`, `th` and `td`.
7. Create `software.html` (list of tools) and `projects.html` (one `article` per sample project).
8. Create `register.html` with a form: `label` and `input` fields of type text, email, tel and date, a `select` list, a `textarea` and a submit button. Add validation attributes such as `required` and `pattern`.
9. Create `contact.html` with the department details.
10. Copy the same navigation links into every page and mark the current page with the `active` class.
11. Start the local server with `python -m http.server 8000` inside the folder and open `http://localhost:8000/index.html`.
12. Test every link, submit the form with empty and invalid values, and resize the browser to mobile width.
13. Check each page with the W3C Nu HTML Checker, fix any reported errors, and record the results in the testing table.

3. Website Program

Save each code section using the filename shown. All pages share one stylesheet and the same navigation links; the link of the current page is highlighted with the `active` class.

3.1 index.html – Home Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Home</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>Department of Master of Computer Application</h1>
  <p>Learn · Develop · Innovate</p>
  <nav>
    <a href="index.html" class="active" aria-current="page">Home</a>
    <a href="courses.html">Courses</a>
    <a href="software.html">Software</a>
    <a href="projects.html">Projects</a>
    <a href="register.html">Register</a>
    <a href="contact.html">Contact</a>
  </nav>
</header>
<main>
  <section>
    <h2>Welcome to the MCA Department</h2>
    <p>The MCA programme focuses on computer applications,
    programming, databases, web development and software
    engineering. Students learn through theory, laboratory
    practice and project-based activities.</p>
  </section>
  <section>
    <h2>Department Highlights</h2>
    <ul>
      <li>Programming and application development</li>
      <li>Database and web technology practice</li>
      <li>Team-based mini projects</li>
      <li>Problem-solving and technical presentations</li>
    </ul>
    <p><a href="courses.html">Explore the courses</a> or
    <a href="projects.html">view sample projects</a>.</p>
  </section>
</main>
<footer><p>2611CSSJ102 – Web Essentials | MCA Department Website</p></footer>
</body>
</html>
```

3.2 courses.html – Courses Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Courses</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>MCA Department</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="courses.html" class="active" aria-current="page">Courses</a>
    <a href="software.html">Software</a>
    <a href="projects.html">Projects</a>
    <a href="register.html">Register</a>
    <a href="contact.html">Contact</a>
  </nav>
</header>
```

```
<main>
  <h2>Representative MCA Courses</h2>
  <p>The following table lists example subjects commonly studied
  in computer application programmes. Actual subjects vary by syllabus.</p>
  <div class="table-wrap">
    <table>
      <caption>Sample Course List</caption>
      <thead>
        <tr><th>Course</th><th>Key Topics</th><th>Practical Work</th></tr>
      </thead>
      <tbody>
        <tr><td>Programming in C</td><td>Variables, decisions, loops, arrays</td><td>Small C programs</td></tr>
        <tr><td>Python Programming</td><td>Functions, collections, files</td><td>Python exercises</td></tr>
        <tr><td>Database Management</td><td>SQL, tables, keys, normalisation</td><td>Database queries</td></tr>
        <tr><td>Web Essentials</td><td>HTML5, CSS, web basics</td><td>Department website</td></tr>
        <tr><td>Data Structures</td><td>Stacks, queues, lists, trees</td><td>Implementation exercises</td></tr>
        <tr><td>Software Engineering</td><td>Requirements, design, testing</td><td>Project documentation</td></tr>
      </tbody>
    </table>
  </div>
</main>
<footer><p>MCA Department Website</p></footer>
</body>
</html>
```

3.3 software.html – Software Tools Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Software Tools</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>MCA Department</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="software.html" class="active" aria-current="page">Software</a>
    <a href="projects.html">Projects</a>
    <a href="register.html">Register</a>
    <a href="contact.html">Contact</a>
  </nav>
</header>
<main>
  <h2>Software and Development Tools</h2>
  <ul>
    <li><strong>Visual Studio Code:</strong> editor for HTML, CSS,
    JavaScript, Python and other languages.</li>
    <li><strong>Python:</strong> language for scripting, automation
    and application development.</li>
    <li><strong>GCC:</strong> compiler used to build C programs.</li>
    <li><strong>MySQL:</strong> relational database system for storing
    and querying structured data.</li>
    <li><strong>Git and GitHub:</strong> version control and code sharing.</li>
    <li><strong>Web browsers:</strong> testing pages and using developer tools.</li>
  </ul>
  <p>Students should use software approved by their institution and
  follow its licensing and lab rules.</p>
</main>
<footer><p>MCA Department Website</p></footer>
</body>
</html>
```

3.4 projects.html – Student Projects Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Projects</title>
```

```
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>MCA Department</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="software.html">Software</a>
    <a href="projects.html" class="active" aria-current="page">Projects</a>
    <a href="register.html">Register</a>
    <a href="contact.html">Contact</a>
  </nav>
</header>
<main>
  <h2>Sample Student Projects</h2>
  <article>
    <h3>1. Student Management System</h3>
    <p>Stores student details and supports searching, updating and
    displaying records. Example tools: Python and MySQL.</p>
  </article>
  <article>
    <h3>2. Online Library System</h3>
    <p>Maintains book records, member details and issue/return status.
    Example tools: HTML, CSS, PHP and MySQL.</p>
  </article>
  <article>
    <h3>3. College Event Registration</h3>
    <p>Allows visitors to view events and submit a registration form.
    Example tools: HTML5, CSS and JavaScript.</p>
  </article>
  <article>
    <h3>4. Expense Tracker</h3>
    <p>Records sample expenses and groups them by category to show
    simple summaries. Example tools: Python and SQLite.</p>
  </article>
</main>
<footer><p>MCA Department Website</p></footer>
</body>
</html>
```

3.5 register.html – Registration Form

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Register</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>MCA Department</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="software.html">Software</a>
    <a href="projects.html">Projects</a>
    <a href="register.html" class="active" aria-current="page">Register</a>
    <a href="contact.html">Contact</a>
  </nav>
</header>
<main>
  <h2>Enquiry / Visit Registration</h2>
  <p>Complete the form below. This sample form demonstrates HTML
  validation; it does not store or send data to a server.</p>
  <form action="#" method="get">
    <label for="name">Full name</label>
    <input id="name" name="name" type="text" required minlength="3"
           autocomplete="name">
    <label for="email">Email address</label>
    <input id="email" name="email" type="email" required
           autocomplete="email">
    <label for="phone">Mobile number</label>
    <input id="phone" name="phone" type="tel" pattern="[0-9]{10}"
```

```
          maxlength="10" placeholder="10-digit number"
          title="Enter exactly 10 digits" autocomplete="tel">
    <label for="date">Preferred visit date</label>
    <input id="date" name="date" type="date">
    <label for="interest">Area of interest</label>
    <select id="interest" name="interest">
      <option value="web">Web Development</option>
      <option value="programming">Programming</option>
      <option value="database">Database Systems</option>
      <option value="projects">Software Projects</option>
    </select>
    <label for="message">Message</label>
    <textarea id="message" name="message" rows="4"></textarea>
    <button type="submit">Submit</button>
  </form>
</main>
<footer><p>MCA Department Website</p></footer>
</body>
</html>
```

3.6 contact.html – Contact Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>MCA Department | Contact</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <h1>MCA Department</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="software.html">Software</a>
    <a href="projects.html">Projects</a>
    <a href="register.html">Register</a>
    <a href="contact.html" class="active" aria-current="page">Contact</a>
  </nav>
</header>
<main>
  <h2>Contact Information</h2>
  <p><strong>Department:</strong> Master of Computer Application (MCA)</p>
  <p><strong>Institution:</strong> SRM TRP Engineering College (An Autonomous Institution)</p>
  <p><strong>City:</strong> Tiruchirappalli</p>
  <p><strong>Address:</strong> [Enter department address]</p>
  <p><strong>Email:</strong> [Enter official department email]</p>
  <p>Replace the bracketed fields with verified college details before
  publishing the website.</p>
</main>
<footer><p>MCA Department Website</p></footer>
</body>
</html>
```

4. CSS Styling (style.css)

Basic CSS provides a reset, consistent colours, a flexbox navigation bar, readable tables, accessible form controls with visible focus, and a small-screen layout using a media query.

```
/* ===== Basic reset ===== */
* { box-sizing: border-box; }
body {
  margin: 0;
  font-family: Arial, Helvetica, sans-serif;
  line-height: 1.6;
  color: #203040;
  background: #f2f6fa;
}
/* ===== Header, navigation, footer ===== */
header, footer {
  background: #15385b;
  color: #ffffff;
  padding: 18px 24px;
}
header h1 { margin: 0 0 4px; font-size: 1.5rem; }
header p  { margin: 0 0 10px; font-style: italic; color: #cfe3f2; }
footer { text-align: center; font-size: 0.9rem; }
footer p { margin: 0; }
nav { display: flex; flex-wrap: wrap; gap: 8px; margin-top: 8px; }
nav a {
  color: #ffffff;
  padding: 6px 12px;
  text-decoration: none;
  border-radius: 4px;
}
nav a:hover, nav a:focus { background: #286b94; }
nav a.active { background: #286b94; font-weight: bold; }
/* ===== Main content ===== */
main {
  max-width: 950px;
  margin: 20px auto;
  padding: 22px;
  background: #ffffff;
  border-radius: 6px;
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.12);
}
h2 { color: #1e5b82; margin-top: 0; }
h3 { color: #15385b; margin: 0 0 4px; }
a  { color: #1e5b82; }
/* ===== Tables ===== */
.table-wrap { overflow-x: auto; }
table { width: 100%; border-collapse: collapse; margin: 16px 0; }
caption { font-weight: bold; padding: 6px 0; text-align: left; }
th, td { border: 1px solid #bdcbd7; padding: 9px; text-align: left; }
th { background: #e2edf5; }
tbody tr:nth-child(even) { background: #f7fafc; }
/* ===== Forms ===== */
label { display: block; margin-top: 12px; font-weight: bold; }
input, select, textarea, button {
  width: 100%;
  padding: 9px;
  margin-top: 4px;
  font: inherit;
  border: 1px solid #bdcbd7;
  border-radius: 4px;
}
input:focus, select:focus, textarea:focus {
  outline: 2px solid #286b94;
  border-color: #286b94;
}
button {
  margin-top: 16px;
  background: #1e5b82;
  color: #ffffff;
  border: 0;
  cursor: pointer;
}
button:hover, button:focus { background: #15385b; }
/* ===== Project cards ===== */
article { border-bottom: 1px solid #d8e1e8; padding: 10px 0; }
```

```
article:last-child { border-bottom: 0; }
/* ===== Small screens ===== */
@media (max-width: 600px) {
  main { margin: 8px; padding: 14px; }
  header, footer { padding: 14px; }
  th, td { padding: 5px; font-size: 0.85rem; }
}
```

4.1 How to Run the Website

- Create a folder named `mca-department`.
- Save all six HTML files and `style.css` in that folder.
- Open a terminal or command prompt in the folder.
- Run: `python -m http.server 8000`
- Open: `http://localhost:8000/index.html`
- Click each navigation link and test the form fields.

If Python is not installed, use a local server such as XAMPP Apache. Opening the files directly also works for this simple site, but a local server is a better way to test website behaviour.

5. Output of the Program

The screenshots below were captured from the website running on `http://localhost:8000`.

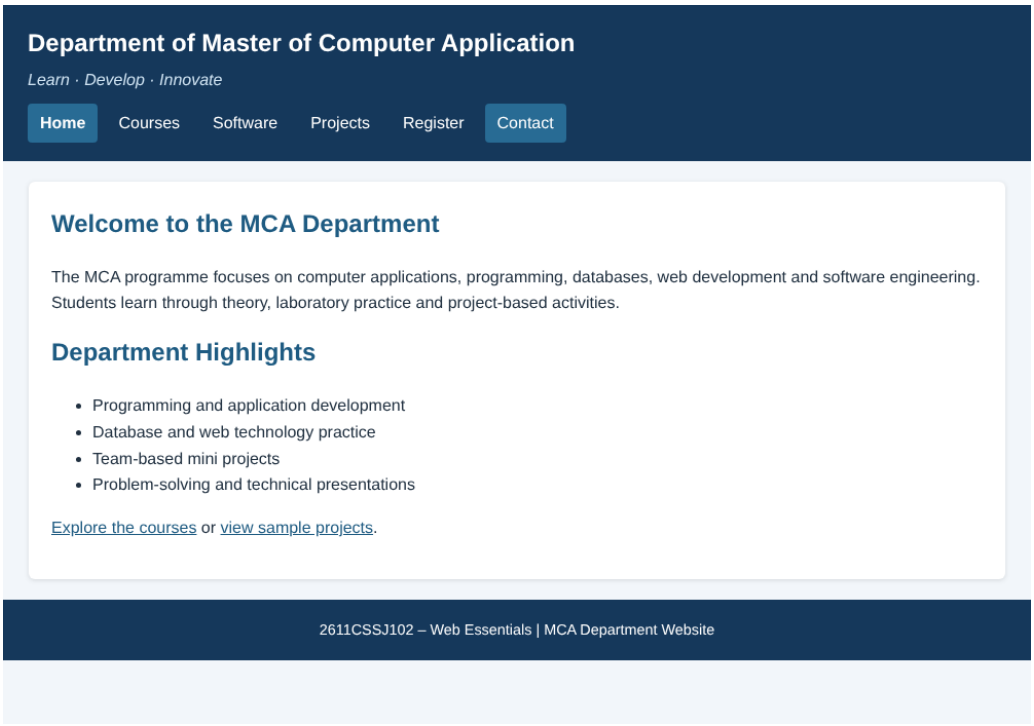


Figure 5.1 – Home page (index.html)

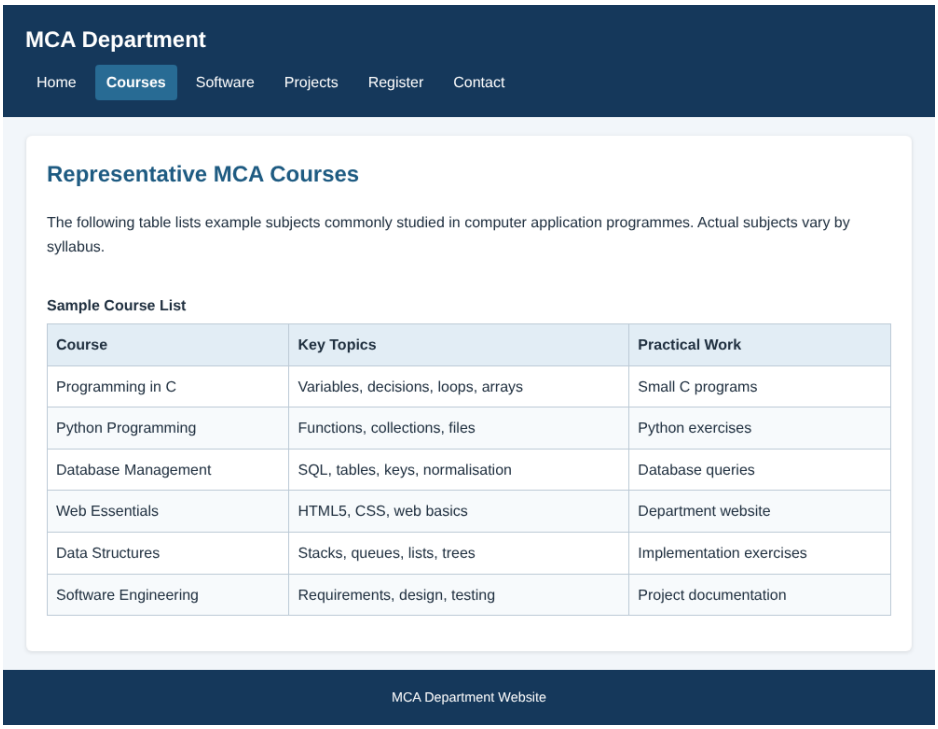


Figure 5.2 – Courses page with the course table (courses.html)

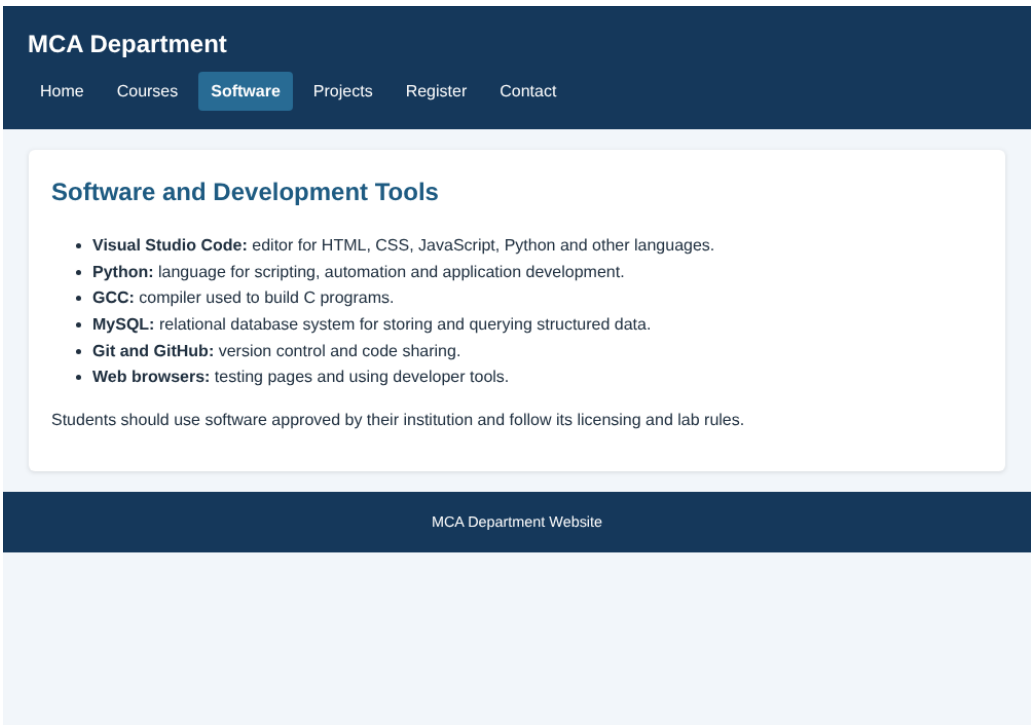


Figure 5.3 – Software tools page (software.html)

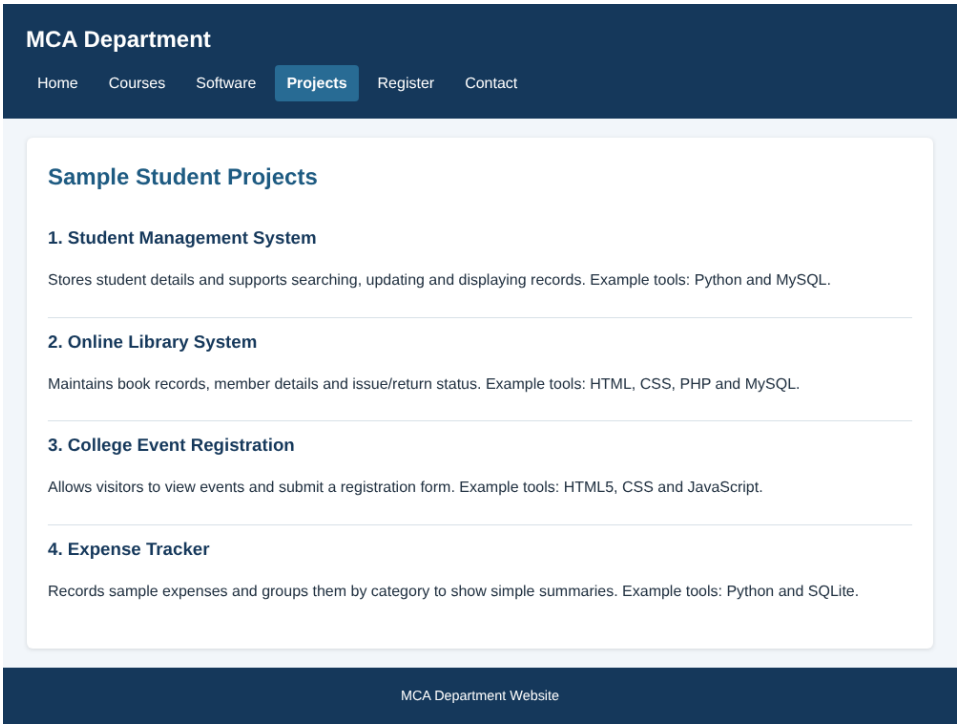


Figure 5.4 – Student projects page (projects.html)

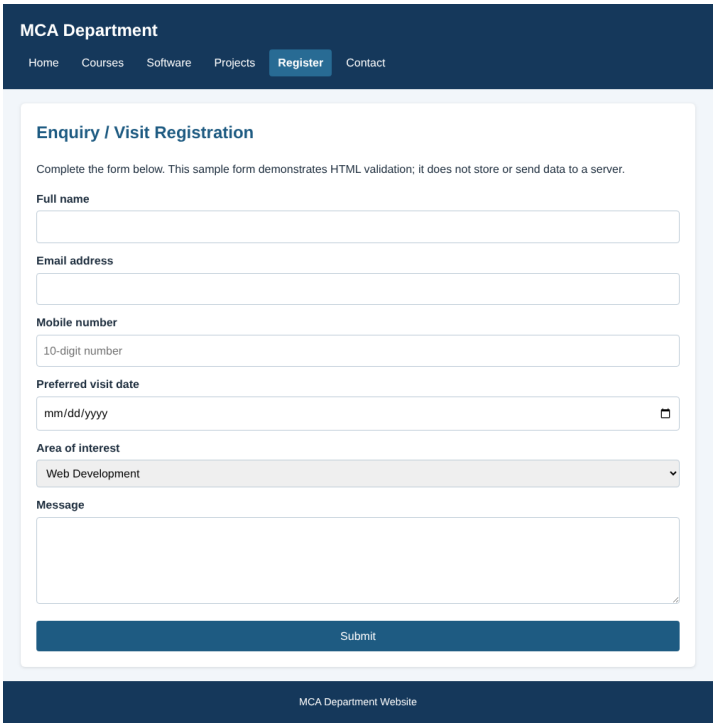


Figure 5.5 – Registration form (register.html)

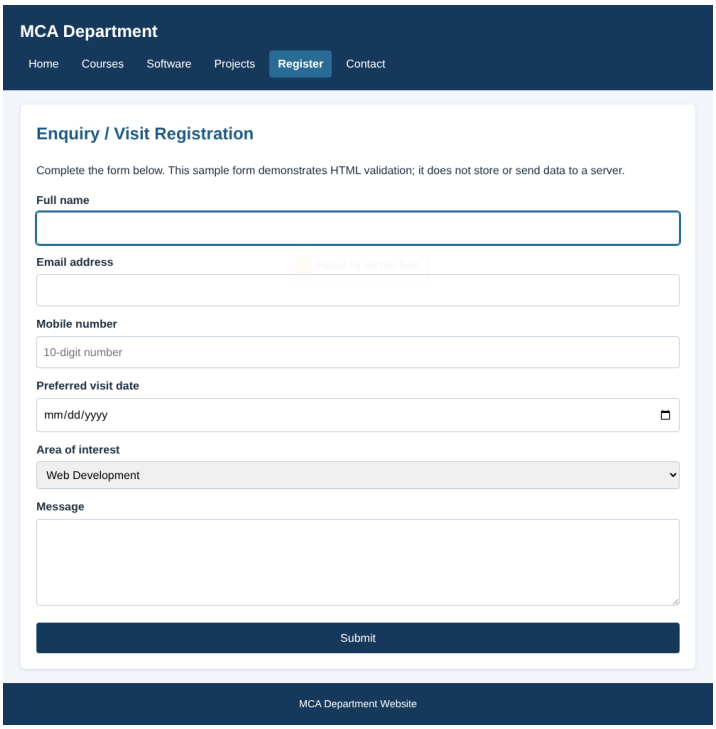


Figure 5.6 – Browser validation after submitting an empty form

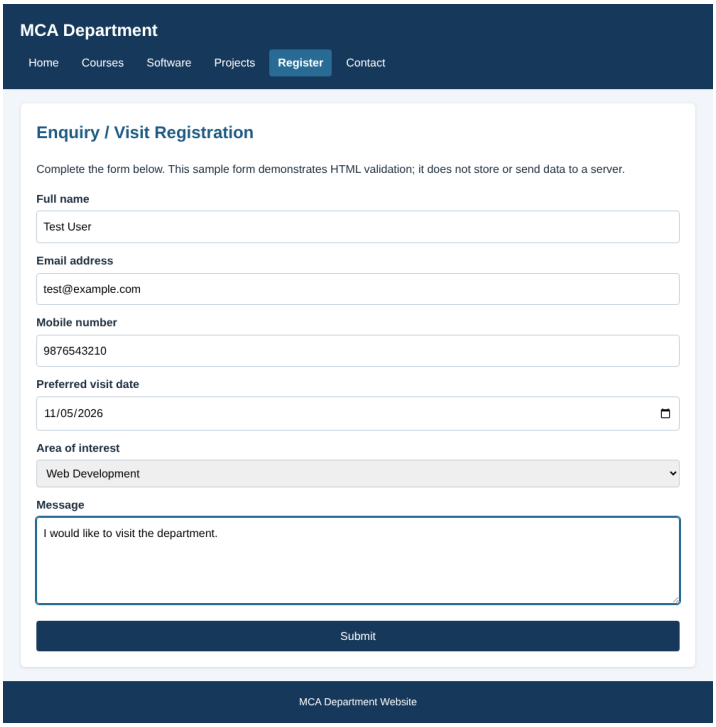


Figure 5.7 – Registration form filled with valid sample data

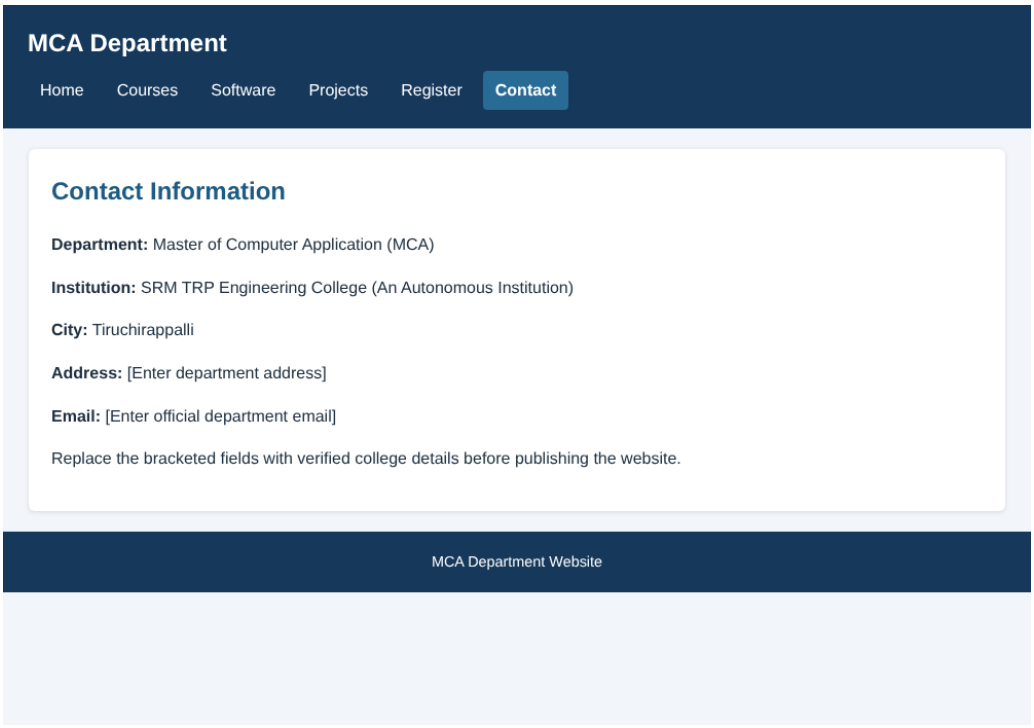


Figure 5.8 – Contact page (contact.html)

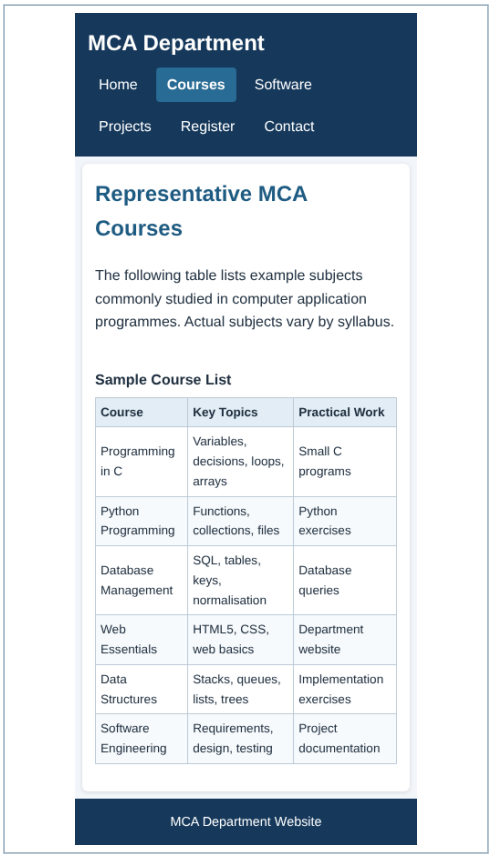


Figure 5.9 – Courses page at mobile width (375 px)

6. Testing and Security

Test ID	Test	Expected Result	Result
T01	Open index.html on localhost	Home page appears	Pass
T02	Click every navigation link	Correct page opens	Pass
T03	Open courses.html	Course table (6 rows) is readable	Pass
T04	Open software and projects pages	6 tools and 4 project descriptions appear	Pass
T05	Submit the form with empty required fields	Browser displays validation prompts	Pass
T06	Enter an invalid email; test phone pattern	Browser rejects invalid email; 10 digits required for phone	Pass
T07	Resize browser to mobile width (375 px)	Content fits with no horizontal scroll	Pass
T08	Check HTML with W3C validator	No reported errors	To be run online

T01–T07 were run automatically in headless Chromium against the local server. T08 needs internet access to https://validator.w3.org/nu/, so it was not run here; paste each page into the checker before submission.

Security Considerations

- Use HTTPS when publishing a real website.
- Do not collect passwords or sensitive information in a demonstration form.
- Client-side validation improves usability but is not a replacement for server-side validation.
- Validate and safely handle all data on a real backend before saving it to a database.
- Use trusted software sources and keep development tools updated.
- Do not publish personal student details without permission.

7. Result and Conclusion

A multi-page MCA department website has been designed using HTML5 and basic CSS. It contains a home page, a representative course list, software tools, sample student projects, a registration form and contact information. The common navigation helps users move between pages, while the stylesheet provides a consistent layout and responsive behaviour.

The aim of the mini project is achieved: the files were saved with the specified names, the site ran on localhost, and the listed tests were completed. Course information and contact details are examples and should be updated to match the actual institution and syllabus.

References

- MDN Web Docs – HTML: https://developer.mozilla.org/en-US/docs/Web/HTML
- MDN Web Docs – CSS: https://developer.mozilla.org/en-US/docs/Web/CSS
- W3C Nu HTML Checker: https://validator.w3.org/nu/